Jiaxuan Chen

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Detailed research curriculum vitae · 19 October 2026

Physics graduate with research experience in high-energy jet simulation, nuclear spectroscopy, thermal transport and X-ray astrophysics. Current AI for Science work spans particle phenomenology, gravitational-wave inference, mathematical physics, statistical identification and machine-learning evaluation. Founder of Omytea, developing personal-agent systems, probabilistic decision tools and motion-analysis products. This research CV includes published work, submitted and unpublished manuscripts, software, research prototypes and exploratory studies, with their respective stages stated below.

AI for Science approach: AI-assisted mathematical exploration, scientific computing, analysis and manuscript development, with project-specific validation and explicit reporting of limitations.

Education

Brown University — Sc.M. in Physics

Providence, Rhode Island, USA · Fall 2024–Spring 2026 · Degree awarded 23 May 2026

- Graduate coursework: Quantum Mechanics; Advanced Quantum Mechanics; Classical Theoretical Physics I and II; Statistical Mechanics; Mathematical Methods for Engineers and Physicists; Group Theory; Electromagnetism and Materials Design.
- Graduate research in the Roloff and Kuehne groups covered CMS jet physics and carbon-nanotube thermal characterization, respectively. These experiences are described individually below.

Shandong University — B.S. in Space Science and Technology

School of Space Science and Physics · 2020–2024 · Degree awarded 21 June 2024

- Undergraduate thesis: *Study of the Outburst Process of the Black-Hole X-ray Binary MAXI J1348-630*. Supervisor: Hongxing Yin.
- Earlier projects included in-beam gamma-ray spectroscopy, axion-like dark-matter phenomenology, stellar-flare spectroscopy and AGN jet formation.

Particle physics, nuclear physics and thermal transport

CMS jet physics — Jennifer Roloff group, Brown University

Graduate research · May–September 2025

- Participated in high-energy experimental-physics research on jet simulation with PYTHIA, studying the connection between a generated hard interaction, parton-shower cascades and the resulting jet structure.

- The project concerned jet topology and observable signatures relevant to beyond-the-Standard-Model searches. The existing research record describes simulated comparison samples for interpreting CMS jet observables.
- Research scope included strong-interaction shower behavior and the interpretation of jet substructure.

Coherent strange-current contributions to X(2370) → ηηη

AI for Science (AI-assisted) · Single-author manuscript · Research: August 2026 · Submitted to Physica Scripta, 9 October 2026

- Examines a proposed prediction for an unmeasured three-pseudoscalar decay channel of X(2370), asking whether the strange-current term included in the ηηη′ channel must also contribute coherently to ηηη.
- Reconstructed light- and strange-current amplitudes from the displayed neutral cubic current, accounting for identical-field differentiation, symmetry factors, FKS mixing and charge-specific three-body phase space.
- At the stated reference inputs, the light and strange amplitudes interfere destructively; the completed ηηη width is smaller by a factor of approximately **30.88**. The charge-complete reference ratio Γ(ηηη)/Γ(KKπ) is approximately **3.315×10^{-4}**.
- Investigated sensitivity to the input parameter box and derived conditional analytic envelopes. The declared sensitivity box contains a cancellation region and is not a joint confidence region; a uniform tenfold suppression over that box is not asserted.
- With an unrestricted overall normalization, the measured-channel fit is unchanged; the correction concerns the unmeasured-channel prediction. It does not establish the resonance’s glueball content, a new experimental signal or a new lattice-QCD result.

Two-photon constraints on the X(2370) region — earlier Belle-data study

AI for Science (AI-assisted) · Phenomenology study · August 2026 · Archived negative result

- Investigated whether a two-photon amplitude analysis could bound the photon coupling of a candidate state in the η(2370)/X(2370) region using Belle data.
- Studied single- and double-pole amplitude families and the identifiability of a photon-width limit. A subsequent reconstruction exposed a Barrelet-type twin ambiguity: the competing amplitudes agreed to approximately **2.5×10^{-14} in $\Delta\chi^2$**.
- The proposed broad-family limit was withdrawn. The retained single-pole range, approximately **5.1–242.7 eV**, did not constrain the predictions used for comparison.
- Preserved the counterexample and closed this line rather than presenting its earlier limit as a physical result. This is distinct from the later, submitted strange-current manuscript.

Strongly supercooled transition dynamics and tachyonic modes

AI for Science (AI-assisted) · Theoretical and computational programme · Research: September 2025 · Latest completed comparison: 16 October 2026

- Investigates bubble evolution and fluctuation dynamics in a strongly supercooled benchmark, connecting an RG-improved potential, O(4) tunnelling bounce, Euclidean mode initialization, analytic continuation, Milne evolution and backreaction.

- The completed conditional calculation follows four trajectories: two thermal-mass branches, each at **pmax = 512 and 1,024**, evolved to **s = 1.11 GeV^{-1}** with 1,799 saved time slices per trajectory. The two branches give field-curve RMSEs normalized by the reference scale w of approximately **14.17%** and **2.10%**.
- Compared shooting and collocation constructions of the bounce, obtaining relative differences below approximately **1.3×10^{-9}** in the escape point and **1.5×10^{-9}** in the action. High-precision reconstruction of the final variance, energy density and pressure from the saved mode coordinates agreed with the working computation to about **3×10^{-13}**.
- The work separates numerical agreement within a conditional model from reproduction of the reference authors' exact implementation. Thermal-branch identification, early-time vacuum preparation, renormalization matching, kinetic terms and quantum self-consistency remain material issues.
- Energy and Ward-identity diagnostics retain nonzero residuals rather than treating stable trajectories as proof of full physical consistency. The calculation does not establish a new particle-production mechanism or observational prediction.
- Current outputs are numerical trajectories, comparison tables and a technical research report; no journal submission for this line is confirmed. Computation and verification were AI-assisted.

Spillway-preheating inverse problems and shape analysis

AI for Science (AI-assisted) · Theoretical research notes · September 2025 · Unpublished

- Studied shape ratios and parameter identifiability associated with a Mansfield–Fan–Lu-type preheating model, using cumulative weighted means, Jacobian-rank analysis and restricted reductions from a two-dimensional problem to a one-dimensional one.
- Developed single-crossing and unimodality arguments within specified parameter windows; counterexamples outside those windows were retained to delimit the statements.
- The resulting theorem notes and interval/numerical workbench do not establish an unrestricted global inverse theorem. Some residual analytic and interval bounds remain open, and successful checker examples do not certify all arithmetic or the full physical reheating dynamics.

Closed-string tachyon representations and order parameters

AI for Science (AI-assisted) · Three related theoretical studies · September 2025 · Unpublished working material

- **Representation dependence:** compared seam-graded/hyperbolic closed-string-field descriptions with oscillator-basis formulations, asking how representation choices affect conditioning and conclusions about a candidate "nothing" vacuum. An earlier broad convergence claim was withdrawn; the surviving representation-conditioned argument remains working material.
- **Typed order-parameter dictionary:** compared closed, winding and thermal tachyon settings through spectral negative-mode flow and renormalized on-shell functionals, with attention to string/black-hole correspondence. The surviving material is a restricted comparison framework, rather than a universal order parameter across all tachyon theories.
- **Curvature-induced phase boundaries:** investigated spectator curvature coupling, exit history and possible dark-matter/gravitational-wave consequences. Candidate witness configurations and finite-difference diagnostics exist, but the required uniform interval, metric and evolution bounds have not closed a general phase-boundary theorem.

- Current status: related theoretical working materials within the same programme; unpublished.

In-beam gamma-ray spectroscopy — Shandong University

Undergraduate research · 2022

- Analyzed gamma spectra from alpha-induced nuclear reactions to identify residual nuclides.
- The existing project record describes energy-matrix gating to isolate relevant spectral features and relate them to reaction products.

Axion-like dark matter — Beijing Normal University research project

Research project · March–May 2022

- Studied axion-like-particle phenomenology through photon survival probabilities and the resulting constraints on candidate parameter space.
- The project linked a theoretical propagation calculation to the interpretation of astronomical signals.

Carbon-nanotube 3ω thermal characterization — Matthias Kuehne group, Brown University

Graduate research · October 2025–February 2026

- Participated in thermal-transport research on single-walled carbon nanotubes using the 3ω characterization framework.
- The associated analytical work addresses finite heater length and environmental coupling: how a heater’s spatial extent, heat capacity and interfaces affect its frequency-dependent thermal response in vacuum, fluid and substrate-supported environments.
- A local manuscript draft is titled *Unified Theory for 3ω Thermal Characterization: Finite Heater Length and Environmental Coupling Effects*. It treats the measured thermal impedance and the assumptions needed to infer material response.

Gravitational-wave inference and astrophysics

Calibration diagnostics for ringdown overtone tests

AI for Science (AI-assisted) · Research · Programme begun June 2024 · Subsequent computational work: 2026 · Working manuscript

- Studies how reference-test dimension, analysis start-time selection, filtering and mean-model conditioning affect apparent evidence for a second black-hole ringdown mode.
- The programme includes historical analysis involving real GWOSC detector noise and a later controlled physical-injection pilot using LALsuite IMRPhenomD waveforms, fixed source parameters, antenna responses and arrival-time delays.
- The physical pilot compares processing and start-time choices with paired H1/L1 noise realizations. Reusing the same noise across analysis arms permits paired comparisons, but does not create additional independent trials.

- A subsequent split-calibration experiment processes candidate 220/221 means through the same observation pipeline, using **199 calibration pairs and 200 separate test pairs**. In its specified synthetic Gaussian-mixture null, the conditioned-mean policy rejected **10/200 test pairs (5.0%)**, with a 95% exact binomial interval of approximately **2.42%–9.00%**.
- A paired raw-mean control rejected **7/200 (3.5%)**. This illustrates that empirical calibration can make a misspecified statistic appear to have a reasonable false-positive rate under one null; false-positive rate alone does not validate the physical mean model.
- The current work does not establish coverage over real detector noise, all nuisance parameters, or an event-level overtone discovery. The manuscript is a substantive working revision of earlier ringdown work, with historical failed approaches retained.

Population-feature significance and finite selection simulations

AI for Science (AI-assisted) · GWTC-3/GWTC-4 study · 2026 · Unpublished manuscript and analysis

- Investigates the effect of finite injection Monte Carlo on population-feature inference, particularly a feature near the secondary-mass edge around **44 solar masses**.
- Uses hierarchical-population tools including gwpopulation, bilby and dynesty, with a GWTC-4 workflow containing **153 events** and approximately **1.066 million injection records**.
- Examines feature-local importance weights, effective sample size and selection variance, rather than relying only on one global Monte Carlo diagnostic.
- The local diagnostic gave an effective sample size of approximately **29,162** above the specified mass edge, compared with a reference threshold of **612**. The initially proposed explanation based on severe selection-Monte-Carlo underconvergence was therefore not supported by that analysis.
- A controlled Beta-null-predictive experiment illustrates finite-simulation error control: at ten injections, the stored experiment reported a naive rejection fraction of **0.259** versus **0.046** for the proposed procedure at nominal 0.05. This is a result for that generative experiment, not a model-independent theorem about the astrophysical feature.
- A draft discusses population-feature certification; the present CV records selection-uncertainty analysis and controlled coverage simulations without asserting a general distribution-free guarantee or a submitted article.

GW250114 nonlinear-mode reanalysis

AI for Science (AI-assisted) · Ringdown programme begun June 2024 · GW250114 posterior reanalysis: June 2026 · Unpublished study

- Reanalyzed seven publicly released posterior sets associated with quadratic ringdown-mode claims, examining analysis start times, amplitude conventions and source-parameter degeneracies.
- The simplified zero-amplitude exclusions ranged from approximately **0.91σ to 3.45σ** across starts. Effective-trials adjustments did not produce the large significance reduction initially hypothesized.
- Examined amplitude correlations with inclination, chirp mass and spin in the released samples, treating these as posterior-geometry diagnostics rather than proof that a physical signal is absent.
- Work uses public inference outputs and does not constitute a new independent full parameter-estimation run, a calibrated global p-value or a discovery of a new quasi-normal mode.

MAXI J1348-630 outburst — Shandong University undergraduate thesis

Undergraduate thesis · September 2023–May 2024 · Supervisor: Hongxing Yin

- Investigated timing and spectral evolution during the outburst of the black-hole X-ray binary MAXI J1348-630 using public **NICER and NuSTAR** observations.
- The thesis describes HEASARC data retrieval, calibration and event filtering with HEASoft/NuSTARDAS; selection of source and background regions; light-curve, spectrum and response extraction; and matching observations across instruments.
- Used **Powspec** for timing/power-density analysis and **XSPEC** for energy-spectrum modelling. The documented reduction includes approximately 4 ms light-curve bins, NuSTAR FPMA/FPMB handling and regrouping spectral channels for fitting.
- Examined NICER observations containing previously identified quasi-periodic oscillations and compared timing features across outburst states. No useful QPO was identified in the examined NuSTAR data; this negative outcome is part of the thesis record.
- Compared relativistic-reflection and simpler absorbed disc/continuum/line models. Joint NICER–NuSTAR fitting exposed differences in temporal overlap and fitted parameters; the thesis discusses substantial residuals and model/data compatibility rather than claiming a uniquely established geometry.
- The work culminated in the undergraduate thesis. Some previously stated outflow interpretations remain model-dependent and are not presented here as newly verified measurements.

Stellar-flare spectroscopy on TESS Objects of Interest

Shandong University · Undergraduate research · April 2022–April 2023

- Studied stellar-flare spectra in systems associated with TESS Objects of Interest, with attention to the radiation environment experienced by potential planets.
- Related stellar activity to questions of exoplanet habitability. Current output: completed undergraduate research experience.

Formation mechanism of blazar jets

Research project associated with University College London · July 2022–January 2023 · Single-author paper published in 2023

- Studied AGN/blazar jet formation and compared the Blandford–Payne and Blandford–Znajek mechanisms through relationships among jet power, accretion-disc luminosity and black-hole parameters.
- The published paper uses existing observations and log-linear fits to discuss where the two mechanisms may be complementary; it is not a claim of discovering either mechanism or measuring a new universal jet law.
- **Publication:** Chen, J. (2023). Formation mechanism of blazar jets. *Theoretical and Natural Science, 19*, 169–175. doi:10.54254/2753-8818/19/20230529.

Astronomical and space-science field training

Shandong University · July 2023

- Completed a six-day field-training programme involving the National Space Science Center, National Astronomical Observatories and Xinglong Observatory.

- The authored training report covers space-weather research, spacecraft-environment effects, a visit to a 50 MeV proton cyclotron, survey astronomy and observational facilities. The programme combined site-based study, technical lectures and a written training report.

Mathematical physics, foundations and discrete mathematics

Local modular convergence for half-line free fermions

AI for Science (AI-assisted) · Single-author mathematical-physics preprint · Research: May–August 2024 · Public version 2, 31 July 2026

- Studies whether compact-local wavelet discretizations recover continuum local modular data for half-line free-fermion systems, and how local convergence differs from global equivalence at fixed cutoff.
- The manuscript treats quasi-free CAR states, faithful block completions, local-state restrictions, modular generators and local modular automorphisms, using operator-theoretic convergence including strong-resolvent convergence.
- Under the stated compact-local wavelet hypotheses, the manuscript proves norm convergence of local-state restrictions and, after faithful block completion, strong-resolvent convergence of modular generators, yielding local modular-automorphism convergence. A locality/passband obstruction delimits raw fixed-cutoff global quasi-equivalence.
- Version 2 narrows an earlier endpoint-rate assertion, distinguishes intervals away from and touching the endpoint, and removes an inapplicable fixed-spacing asymptotic formula. The amended scope is retained rather than repeating the older broader claim.
- **Preprint:** *Convergence of local modular data for half-line free fermions and a locality/passband obstruction to raw fixed-cutoff global quasi-equivalence*. Zenodo, doi:10.5281/zenodo.21717255.

Digit dependence under translation-conjugated cyclic rotation

AI for Science (AI-assisted) · Single-author discrete-mathematics manuscript · Research: May–August 2024 · Unpublished

- Determines how conjugating a cyclic digit rotation by modular translation changes the essential dependence of each output digit on its inputs in a fixed-radix register.
- Classifies the complete pivotality matrix in terms of the divisibility order of the translation: for nonzero translation c in a base-a, k-digit register, the largest pivotal-input count is **$k - v_a(c)$**. Dense dependence support is distinguished from the probability that a random input actually changes an output.
- Gives a **$\Theta(\log k)$** minimum-depth separation for the increment conjugate versus wiring-only rotation in the stated bounded-fan-in model. Exact coordinate-change counts use paired-state finite automata, with **$O(k^2a + ka^2)$** integer-arithmetic operations for the general translation/rotation-distance algorithm.
- Reproducibility material includes direct definition-level oracles, separate paired-transition implementations and integer-arithmetic checks. The recorded finite grid covers **2,880 parameter cases and 75,504 matrix entries**; these checks support implementation validation rather than replace general proofs.
- The manuscript explicitly distinguishes essential dependence, average sensitivity, circuit depth and physical locality, and acknowledges that finite-state difference methods have prior literature.

Locality, short descriptions and machine-dependent ranking

AI for Science (AI-assisted) · Single-author foundations/discrete-mathematics manuscript · Research: May–August 2024 · Manuscript preparation: September 2026 · In preparation

- Uses the finite-register rotation/conjugation example to test whether short descriptions or a universal prior can provide a machine-independent preference for one locality structure.
- Studies conditional shortest-prefix-code weights and bounds the description-length difference between the two rule families independently of register size, leaving machine-dependent constants in their prior odds.
- Constructs universal prefix-machine choices that reverse the ordering. The scope excludes a machine-independent ranking for this pair; it does not rule out every locality selector, every Born-rule derivation or all forms of physical emergence.
- An explanatory companion to the digit-dependence technical paper, developed as a separate manuscript.

Observer-state foundations programme — Random Sequence Scintillation

AI for Science (AI-assisted) · Foundational research and unpublished book manuscript · Research: May–August 2024

- Explores an observer-state ontology and a transition framework weighted by an algorithmic prior, asking which additional assumptions are required to recover locality, modular dynamics or quantum-probability structure.
- An earlier public preprint, *A Universal Prior Cannot by Itself Select a Preferred Factorization: A Bounded-Alphabet Locality No-Go, with a Route-by-Route Map of the Same Gap in Born-Rule Derivations*, is recorded at doi:10.5281/zenodo.21325099. Its later finite-register descendants are identified separately above.
- Additional working projects include *What an Algorithmic-Idealism Modular Identification Requires: A Foundational Delimitation* and *Boltzmann Brains under an Observer-State Ontology*. Both remain unpublished working material.
- Further directions include *Other Minds under an Observer-State Ontology* (in preparation) and a paused Born-rule/naturalness study.
- The associated unpublished book manuscript, *Sovereign Scintillation Over Grand Fluctuation: An Observer-State Theory of Physics*, records the broader framework and its open problems.

Machine learning, calibration and statistical identification

Calibrated belief-state probing in language models

AI for Science (AI-assisted) · ML-interpretability research · 2026 · Exploratory real-model results, unpublished

- Asks whether internal representations contain useful information about answer correctness beyond a suitably controlled output baseline, and whether such scores support abstention with explicitly defined risk targets.
- The current real-model pilot uses **Qwen2.5-0.5B-Instruct on ARC-Challenge**, with 1,116 training, 295 calibration and 1,165 test questions. It preserves 2,576 question-level records and compares output summaries, middle/final dense representations and a permuted-label control.

- The main comparison, output plus middle-layer features versus output alone, produced an AUROC difference of **0.00218**, with a paired 95% bootstrap interval of approximately **−0.01857 to 0.02283**. This does not show a clear incremental gain and does not prove equivalence or absence of internal information.
- Separated representation quality from calibration construction. A classical order-statistic construction produced an admissible policy for the output baseline where the original threshold grid certified none. On the test set that policy answered **304/1,165** questions, with **98** answered incorrectly; marginal error and error conditional on answering are kept distinct.
- A subsequent frozen six-representation, train-only development comparison used nested cross-validation and PCA/random-projection controls. Middle-layer additions failed the prespecified continuation rule, so expansion of that small-model linear branch stopped.
- The results concern one model, task and frozen workflow. They are not a Gemma/SAE replication, evidence of a model's subjective beliefs, a causal intervention result or a deployment guarantee. The project uses AI-assisted implementation and analysis.

Auditing expected-correctness and calibration claims

AI for Science (AI-assisted) · Mathematical diagnostics associated with the probing project · 2026 · Unpublished

- Examines the distinction between marginal expected correctness, acceptance-conditioned correctness and guarantees conditional on a realized calibration sample.
- Constructed finite probability examples showing that particular printed lower bounds can exceed their stated target when the procedure's participation and conditioning are represented explicitly.
- Analyzed final-hidden-state/output-head relations and exact stored-head rank to avoid treating an information-equivalent final-layer coordinate as additional evidence beyond complete outputs.
- The work challenges specific mathematical statements and baseline choices; it does not refute all empirical scoring methods, prove that internal representations are useless or claim a published correction.

Dependence models and exact identification under uniform single deletion

AI for Science (AI-assisted) · Single-author mathematical-statistics working manuscript · October 2026 · Unpublished

- Studies the identification of the mean mark of the minimum-score record when one uniformly chosen record is deleted, even if the entire remaining observation law is known exactly.
- Separates arbitrary finitely exchangeable populations, mixtures of iid records and a single iid law. It also distinguishes observations retaining complete numerical scores from observations retaining only the ordered binary mark word.
- The manuscript gives different ambiguity diameters across those models. For the complete-score experiment, arbitrary populations attain the central-binomial reciprocal as a supremum, common-latent iid mixtures give **N/4^(N−1)** as a supremum, and one iid law is identifiable.
- Uses finite-channel duality, positive-population constructions, polynomial approximation, switching identities, polarization and score-domain lifting. The argument explicitly identifies classical components instead of presenting the deletion operator or generic convex duality as a new invention.

- Related finite-extendibility work studies how a finite source length approaches the iid-mixture endpoint, and when complete word laws contain more target information than rank means. Exact finite rational certificates accompany the proofs.
- These remain mathematical working results. Historical priority, research significance and external scientific review remain unresolved; no finite-sample model-performance advantage is asserted by the identification formulas alone.

Full observations versus rank-summary statistics

AI for Science (AI-assisted) · Mathematical-statistics subproject · October 2026 · Unpublished

- Investigates information lost when a full deleted-word observation is compressed to rank means or to a final coordinate sum, rather than assuming these summaries are sufficient.
- Developed exact finite-source examples and a symmetric-channel minimax formulation, separating the risk available from full distributions from that available through additive summaries.
- A fixed 9-to-4 study establishes a strict asymptotic worst-case-risk separation; a conservative finite-sample sufficient bound of **50,000 independent whole experiments** appears in a separately checked supplement.
- The project treats the definition of the observation, the allowed source class and precision of a real-valued summary as mathematical assumptions. It does not claim a general memory advantage, universal compression impossibility or practical sample-complexity improvement for LLM deployment.

Positive population ambiguity: composition crossover and critical growth

AI for Science (AI-assisted) · Single-author mathematical research manuscript · 19 October 2026 · Unpublished

- Studies positive populations on fixed-weight binary words whose complete laws after one uniformly indexed deletion coincide, imposing a separate chi-square-radius cap on each population.
- Derives a normalized ambiguity profile across sparse, critical and Gaussian composition regimes. The critical formulation includes compensated marked-Poisson components with odd signed-slot parity, rather than restricting the optimization to one scalar projection.
- For each fixed positive critical intensity, the current manuscript obtains exact large-radius growth proportional to **$\log \kappa / \log\log \kappa$**, with a matching coefficient from a feasible positive lower construction and a spatial-first-moment entropy upper bound.
- The analysis shows why convergence on bounded radius intervals cannot automatically be extended over the entire half-line of radii: the finite-intensity profile can exceed its Gaussian endpoint by an unbounded factor at large radius.
- Supporting work includes full cubic/extremizer structure, finite-trace positive constraints and Gaussian/Poisson endpoint comparisons. These are connected components of one research programme, rather than a separate publication for every intermediate draft.
- The latest 23-page manuscript incorporates proof clarifications and classical attribution. Its limits include even source lengths, a word-length limit taken before the large-radius limit, and no general simultaneous finite-source/growing-radius or empirical sample-complexity theorem. External significance and publication review remain open.

Condensed-matter and stochastic-dynamics studies

Time-crystal signal discrimination

AI for Science (AI-assisted) · Controlled small-system numerical study · July 2026 · Unpublished

- Investigates whether an apparent Floquet period-doubled signal can be distinguished from a noise-driven mimic rather than inferred solely from an oscillating trace.
- Uses small systems of approximately 9–14 sites, a coherent pi-mode comparison and shared-ancilla noise, with time-window, shot-count and calibration/validation controls.
- An initially marginal result was traced to trajectory/calibration randomness. A prespecified resolving experiment across five ensembles supported continuation of the numerical discriminator, with later windows giving more stable separation.
- The finding concerns the specified finite-system experiment. It is not a new observed time-crystal phase, an experimental refutation of published data or a completed journal article.

Finite-sample distinguishability of collective criticality

AI for Science (AI-assisted) · Theoretical deep-dive study · June 2026 · Unpublished notes

- Studies how finite observations distinguish a critical system from a near-critical alternative using Gaussian/Ornstein–Uhlenbeck lattice models, finite-size scaling and critical slowing down.
- Uses information-theoretic distance and testing bounds, including KL/Le Cam arguments, to examine how a restricted sample requirement changes with distance from criticality.
- Existing statistical/econometric precedents limit the originality of broad versions of the argument. Current material is a scoped analysis and literature comparison, without an accepted new universal impossibility theorem.

Calibration and dissipation in filtering models

AI for Science (AI-assisted) · Controlled stochastic-model study · June 2026 · Unpublished

- Compares prediction calibration and dissipation proxies in a binary telegraph hidden-Markov model with exponential filters, using Brier/ECE measures, nonpredictive information and coarse entropy-production quantities.
- The stored simulation spans 200,000 steps. Symmetric matched settings favored the same policy, while asymmetric or misspecified settings separated the objectives.
- The direction of the dissipation difference changed across settings. This rejected the strong working claim that improved calibration must invariably cost more dissipation; it does not establish a general thermodynamic law from a simulation proxy.

Further physics-research directions

Research dossiers and preregistration drafts · 2026

- Additional preparation includes a gravitational-wave ringdown census for a future catalogue, dark-energy evidence/analysis questions, biosignature certifiability and broader time-crystal source/data surveys.
- These remain scoped research directions rather than completed experimental studies or independent publications.

Physics-ML benchmarking and scientific evaluation — The Well

Denominator conditioning of rollout VRMSE

AI for Science (AI-assisted) · Benchmark-forensics research · Research: July 2026 · Subsequent analysis and revisions through October 2026 · Working manuscript

- Studies how an unstated denominator floor, normalization space, time window and aggregation convention affect reported physics-simulation rollout errors.
- Examines Rayleigh–Taylor trajectories, sampled Rayleigh–Bénard data and a 17-dataset data-only screen, with checkpoint reevaluation involving U-Net, FNO, TFNO and ConvNeXt families.
- For one fixed UNetClassic checkpoint evaluated in **raw physical units over rollout steps 6–12**, with fixed predictions and aggregation, changing only the floor from 10^{-7} to 10^{-5} changed a selected-window rollout VRMSE from approximately **16.46 to 1.93**. This is metric sensitivity, not improved predictions.
- Reconstructs model-specific coverage and retains divergent/error-heavy runs instead of silently excluding them. Local checks cannot fully recover historical checkpoint weights, scoring space and all publication-table conventions, so the study is not labelled an exact reproduction of every historical score.
- Manuscript: *An Unstated Constant Inside a Benchmark's Metric: Denominator Conditioning of Rollout VRMSE on a Physics-ML Simulation Benchmark*. A TeX/PDF and reproducibility package exist; no current journal-submission receipt or completed scientific-acceptance process is confirmed.

Active-matter non-identification and offline metric tooling

AI for Science (AI-assisted) · Benchmark subprojects · July 2026 · Unpublished analysis and local tooling

- Reconstructed available sampled active-matter field frames and quantified floor-dominated denominator summaries. The observed sample is partial rather than a complete dataset census.
- Constructed realizable synthetic error fields with identical denominator summaries but different ranking behavior, demonstrating that denominator summaries alone do not identify checkpoint ranking changes.
- This is an information-limit/counterexample result, not an observed reversal of real model rankings. Paired real checkpoint errors and an independently consequential scientific example remain missing.
- Built a data-variance screening and offline regrouping workflow from packaged per-window scalars, with fixtures and reproducible numerical tables. Packaging checks are kept separate from scientific validation of the manuscript.

Public upstream contributions to The Well

AI for Science (AI-assisted) · Open-source contributions · Research begun July 2026 · Merged into upstream master

- **Pull request #92, merged 23 July 2026:** propagated the stabilizing epsilon parameter through VRMSE/VMSE metric paths.
- **Pull request #93, merged 12 October 2026:** changed long-horizon validation time curves to aggregate across batches rather than retain only the final batch.

- Claude Code assistance is disclosed in the pull request. These changes correct specified metric behavior; they do not independently demonstrate improved scientific forecasts, sample-weighted aggregation in every path or inclusion in a particular released package.

Additional benchmark-method research

Historical plans and partial research assets · July 2026

- The wider programme includes anytime-valid/e-process rollout certification and related detector/fixture designs. Their current complete manuscripts, final certificates and submission receipts are not established in the available restored source tree.
- These lines are retained as partial research scope, rather than counted as completed benchmark products or publications. Earlier exploratory branches that were reduced or discontinued are not represented as active experiments.

Entrepreneurship and personal-agent systems — Omytea / NYTH

Founder, Omytea

Products and independent research umbrella · 2026–present · omyteaai.com

- Founded Omytea as the umbrella for personal-agent products and scientific research. NYTH is the current principal agent product; probabilistic prediction and world-model research remain a distinct, continuing line.
- Defines product direction and coordinates AI-assisted research, implementation and verification across persistent private experience, permissioned communication, discovery, collaboration, scientific work, capability exchange and controlled interactions with tools and devices.
- The broader direction connects people and persistent personal agents, combining each agent’s private experience and continuing identity with revisable user context, relationships and explicitly authorized action.
- Current stage: partially implemented product, with local workflows and controlled evaluations.

Persistent personal agents, identity and relationship permissions

NYTH product architecture and implementation · 2026 · In development

- Coordinates durable identity and account structures, separate human/agent identities, relationship permissions and revocation, with storage and recovery behavior rather than only an ephemeral conversation demo.
- The private-experience layer supports records, revisions, withdrawal and provenance, connecting a personal home interface to an agent’s continuing context.
- Expression authority separates the owner’s permission to express information from the receiving party’s rules and the execution mechanism. Its design considers relationship, purpose, source, expiry and quota boundaries.
- Model-level intent following is evaluated separately from deterministic permission plumbing.

Selected-record provenance and continuity

Implemented local workflow · October 2026 · Controlled validation

- Coordinated AI-assisted development and verification of a private preview-and-selection workflow for importing explicitly chosen records, with deduplication, source revisions, watermarking and source inspection.
- Withdrawal and owner edits are propagated so that superseded or removed content is not silently revived through recovery or a second import.
- A controlled desktop-browser path demonstrated that records remain unselected by default, one chosen record can be stored and its source revisited, and unselected text is not retained in the temporary test database.
- This is a selected-source interchange workflow, not a claim of complete native imports from every provider, access to hidden memories or automatic synchronization of all past conversations. Local regression failures elsewhere remain open.

Human–agent communication and model intent-following

Product research and bounded model evaluation · 2026

- Studies messaging among people, people and agents, and agents acting through relationships, with distinct receiving, delegation, clarification and expression policies.
- Evaluated local language models on positive and negative intent-following cases, including explicit prohibitions against sending a message.
- A tested configuration failed a predeclared all-cases acceptance rule after selecting a reply despite a current prohibition. It was not adopted for automatic expression selection; no actual third-party delivery followed that evaluation.
- This negative result distinguishes valid output format and deterministic permissions from reliable semantic compliance for the tested configuration.

Capability discovery, service exchange and task artifacts

NYTH product development · 2026 · Partial implementation

- Develops discovery and collaboration around task inputs, work artifacts, capability descriptions, service quotations, results and follow-up, rather than treating a model response as a completed service.
- Existing bounded workflows include a free quotation-audit service, a capability shelf and permissioned artifact access, alongside design work for creator experience and capability exchange.
- Commercial settlement, payments and refunds remain in the product design. Current implementation centers on bounded free-service workflows and authorization.
- Tool and OpenAPI work-package interfaces preserve authorization and result provenance through bounded task and artifact contracts.

Scientific intelligence and controlled work environments

Research and product architecture · 2026

- Investigates how personal agents can perform scientific tasks through source-grounded research, explicit work packages, tool interfaces and inspectable artifacts.

- Explores persistent research context, clarification, task decomposition, fixed scientific control cases and the distinction between a runnable workflow and a valid scientific result.
- Autolab, robotics and embodied scientific work are longer-term directions; present work focuses on computational research workflows and fixed control tasks.

Prediction and world-model research

Continuing Omytea research line · 2026 · Unpublished research and prototypes

- Explores probabilistic representations of multiple possible futures, evidence-driven updates, proper scoring and auditability of forecasts. The original programme uses density-matrix/open-system notation, while the current programme allows whichever methods fit a task.
- Investigates the distinction between state bookkeeping, probability-format validity, numerical scoring correctness and forecast accuracy on resolved outcomes.
- Historical-question workflows produced valid probability outputs in controlled comparisons; scoring implementations also have finite-input agreement checks. Neither establishes predictive superiority or a new world model with general real-world accuracy.
- An open-system extension has completed development computations with a negative research-gate outcome; confirmatory work and a submission-ready paper are not established. Negative results remain part of the programme rather than being promoted to a model advantage.

Omytea Quantum Substrate

Open-source numerical research library · 2026 · Apache-2.0

- A dependency-light standalone Python library using NumPy and the standard library for sparse hypothesis grids, joint-entity hypotheses and density-matrix/open-system bookkeeping.
- Includes representations of off-diagonal correlation and Lindblad-style evolution as a formalism for classical inference and probabilistic state updates.
- Existing project records identify a PyPI release line and Zenodo concept DOI 10.5281/zenodo.20293619. Release infrastructure uses GitHub OIDC trusted publishing.
- This is classical numerical computation using quantum-information mathematics. It is not a physical quantum computer, a demonstrated quantum speedup or evidence that the forecast system outperforms strong classical methods.

Personal Future Console

Decision-support software prototype · 2026

- A personal decision interface for expressing questions about uncertain futures, representing scenarios and probabilities, updating with evidence and displaying coherence.
- The existing prototype includes text input and video/webcam interfaces, with local-model and optional real-time-video components.
- The interface and state-update mechanisms are separate from empirical calibration. Current records do not establish a production camera workflow, sustained real users or predictive improvement caused by a “quantum” representation.

GolfCal-1k

Benchmark specification and synthetic baseline study · 2026 · In preparation

- Specifies a calibration-oriented benchmark for golf-shot forecasts, using landing-zone distributions, Brier and energy scores, reliability diagnostics, adaptive-conformal targets and strokes-gained questions.
- Defines data schema and reference baselines, including hierarchical Dirichlet-categorical and neural approaches. A stored development snapshot contains **414 synthetic shots**; “1k” is the target benchmark scale rather than a claim of 1,000 real recorded shots.
- The **414-shot development snapshot** did not clear its principal gates; its coverage and strokes-gained columns used a placeholder and a proxy, respectively. The CV therefore records specification, synthetic evaluation and method design, without claiming real adaptive-conformal coverage or demonstrated on-course benefit.

Devices, compute and commercial-system research

Product and feasibility research · 2026

- Explores personal agents embedded in everyday devices and furniture, linking device observations to private inputs and controlled action interfaces.
- Research includes native/local compute, hardware availability, licensed routing, service-cost scenarios, supply chains, prior art and different disclosure/release choices.
- Related work examines wireless identity and authorization mechanisms and their failure cases. A mechanism analysis or offline input chain does not establish a working base-station security product, a deployed household device or validated physical actuation.
- Geographic discovery, feeds, matching/judging and demanding technical-use cases remain in the full product scope, with implementation maturity tracked by specific workflows rather than presented as one completed platform.

Motion analysis, sensing, visualization and software

Swindo — monocular golf-motion reconstruction and local Studio

Product engineering and geometric analysis · 2026 · Prototype in development

- Develops a one-phone-video workflow for golf-motion analysis, combining image pose estimates, an external height/profile reference and constrained fitting to a motion representation.
- The current Studio supports pose extraction, fixed-link profile fitting, playback/raw-data switching and JSON export. Desktop and narrow-screen workflows have been exercised locally with an authorized video.
- The research examines identifiability limits of monocular reconstruction: depth and absolute scale are not determined from a single image stream without assumptions; unsynchronized views cannot simply be treated as stereo.
- The displayed clubhead path is inferred from arm geometry rather than directly detected, and face-angle interpretation is limited by temporal sampling and rotation between frames.

- Retains noisy-corner rejection and failing cases rather than weakening them to obtain a clean test summary. Current local UI/CLI validation does not establish centimetre-level ground truth, complete dynamic-camera robustness or field-ready biomechanical accuracy.

Swindo IMU hardware, timing and measurement integrity

Embedded-sensing prototype and diagnostic research · 2026

- Investigates a motion-sensor path involving nRF52840 and STEVAL/LSM6DSV320X-class hardware, separating product firmware from bench diagnostics.
- Work addresses SPI identity checks, register modes, data-ready clock domains, synchronization, sample ABI and interrupt timing, with host-side and boundary-condition fixtures.
- A recorded physical identity read did not match the expected sensor identity and yielded no valid motion samples. That unresolved result is retained; later offline firmware work is not described as a successful new physical capture.
- Static timing analysis also identified a candidate lower bound exceeding its stated cycle budget. This is an analytic timing counterexample, not a measured runtime benchmark or proof of field readiness.

WiFi human-sensing feasibility and data auditing

Comparative research and offline validation tooling · October 2026

- Evaluates how real channel-state information, RSSI scans, DSP, heuristic interfaces, trained pose estimators, public weights and synthetic demonstrations support different kinds of sensing claims.
- Examines benchmark split, coordinate-system, denominator and threshold choices, including MM-Fi-related evidence, and compares RF approaches with Swindo’s visual/IMU paths.
- Produced a source-traced offline audit and proposed synchronized ground truth, held-out people/rooms and time-shuffled RF controls for evaluating a genuine sensing contribution.
- No owned live-RF pose accuracy, clinical physiological measurement, through-wall golf reconstruction or independently trained WiFi model is claimed. Third-party benchmark scores are not presented as personal results.

Scientific personal-site visualization and celestial instruments

Academic portfolio and interactive scientific visualization · 2026

- Develops a bilingual academic portfolio and interactive astronomical views, including a sky/galaxy scene, Solar-System scale representations and geographic research/travel visualization.
- The Nye Clock is a celestial sexagenary instrument connecting a Sun–Earth–Moon orrery with a traditional calendar/natal-chart interface.
- Emphasizes catalogue provenance, scale fidelity, rendering performance and the limits imposed by incomplete or unevenly distributed data.

Agent-runtime and technical-system comparison studies

Technical research documents · 2026

- Comparative work examines provider/tool registries, session state, side effects, read-only skill channels, data sovereignty, persistent-agent runtime design and verification boundaries.

- Includes source-based analyses of agent networks, execution harnesses and frontier-model ecosystems, translating observed behavior into proposed local architecture and tests.
- External-system features are distinguished from personally implemented functionality; research recommendations are not counted as delivered product behavior merely because they appear in a report.

Technology policy, political economy and books

What Public AI Compute Records Can Establish: Evidence, Access and Comparison

Single-author policy-research manuscript · Research: January–July 2026 · Submitted 7 October 2026

- A documentary audit of **six purposively selected public AI-compute programmes**, asking what public records can establish about resource allocation, actual consumption, outputs and policy effects.
- Fixes the perspective of an unauthenticated public observer and a dated source boundary. The Japanese ABCI case distinguishes activities, participants and billing groups, rather than treating them as interchangeable units.
- Uses rate/unit counterexamples to show why equal accounting points do not necessarily imply equal GPU device-hours. Develops reporting constraints around unit crosswalks, denominators, resource definitions and comparison access.
- Within the defined public-source scope, the study did not identify a qualifying named-project consumption linkage. That finding does not imply the internal records do not exist, that the programmes failed or that a causal policy effect has been estimated.
- Submitted to the *International Journal of Technology, Policy and Management*, with a 24-page main manuscript and 33-page supplement. The last confirmed journal stage is **Screening**, not confirmed external review or acceptance.

Inference Rent in Generative AI: Intellectual Monopoly and Metered Access to Non-Rival Capability

Single-author working paper · Research: January–July 2026 · Public SSRN preprint

- Examines how high fixed-cost model training, scalable inference and revocable metered access can turn non-rival model capability into priced scarcity.
- Situates “inference rent” alongside information-goods theory, intellectual monopoly and platform/data rentiership, distinguishing above-normal returns from service costs and from all AI revenue as a whole.
- States scope conditions, objections, potential falsifiers and an empirical research agenda. It is a conceptual political-economy argument, rather than an estimate that every AI firm earns monopoly rent.
- SSRN abstract 6950759; revised public record confirmed in the September 2026 archive.

Frontier AI and the Limits of Monopoly Self-Correction: Inference Rent, Compute Bottlenecks, and Platform Lock-In

Single-author working paper · Research: January–July 2026 · Public SSRN preprint

- Reconstructs three routes through which monopoly might correct itself: reproduction/entry, contestable-market competition and Schumpeterian displacement.
- Analyzes conditions under which concentrated compute, model control, enterprise integration, feedback and switching costs weaken those mechanisms.
- Addresses countervailing forces including open weights, price declines, hardware commoditization, efficiency shocks, regulation, multihoming and demand saturation.
- The argument identifies conditional mechanisms rather than proving permanent monopoly or estimating persistence. SSRN abstract 6950761.

Camera, Player, Engine: An Epistemology for Writing About a System You Are Inside

Single-author methodological essay · Research: January–July 2026 · Public SSRN preprint

- Distinguishes descriptive, strategic and operational positions when the analyst also participates in constructing the technological and political system under discussion.
- Develops a practical distinction between what is known and what is wagered, illustrated through claims about AI rents and their persistence.
- A methodological framework for claim discipline and reflexive analysis. SSRN abstract 6950818.

An Invitation After Abundance: Reading the Present, Wagering the Future

Single-author English-language book · Research and writing: January–July 2026 · Self-published, 2026

- A long-form political-economy study of AI, quantum technology, brain–computer interfaces, gene editing, robotics and fusion, asking how expanding productive capacity interacts with ownership, institutional distribution and private control.
- Introduces technological-monopoly-rent questions within a wider account of simultaneous technological change, combining interpretation of the present with explicit judgments about possible futures.
- Existing production materials include PDF, EPUB and an editable manuscript; the local finished English proof contains 1,162 pages.
- ISBNs: **979-8-9960105-0-9** (ebook), **979-8-9960105-2-3** (paperback), **979-8-9960105-3-0** (hardcover).

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Single-author Chinese-language book · Research and writing: January–July 2026 · Self-published, 2026

- A parallel original political-economy work addressing technological change, social institutions and the distribution of gains from productive capacity in a Chinese-language context.
- The English and Chinese books are parallel original works.
- ISBN **979-8-9960105-1-6** (hardcover).

Essays in development

Unpublished writing projects · Research: January–July 2026

- *The Abundance That Isn't* examines technological abundance and the institutions through which access may remain scarce.
- *The Rent Machine* develops a rent-decomposition account of that enclosure. These are standalone essay drafts; no new magazine acceptance or publication is claimed.

Exploratory scientific and entrepreneurial research

Artificial cells and cell-free systems

Scientific and commercial-feasibility research · 2026 · Research dossier

- Maps distinctions among cell mimics, protocells, minimal cells, cell-free systems and autonomous bottom-up cells, identifying where scientific demonstrations differ from product claims.
- Compares value chains and candidate opportunities in measurement, quality control and interoperability, with explicit propositions awaiting evidence.
- Current stage: scientific desk research and venture exploration, before wet-laboratory prototyping or customer validation.

Cross-domain technology strategy

Research dossiers and comparative frameworks · 2026

- Compares energy, computation, space, intelligence and life-science pathways through physical constraints, technical evidence, competitive structure and possible points of entrepreneurial intervention.
- Treats feasibility, research significance and commercial demand as different questions, preserving disconfirming evidence rather than turning every technology interest into a completed venture.
- Current outputs are comparative research dossiers and candidate venture questions, with physical feasibility and demand validation treated separately.

Publication and manuscript index

The index groups scholarly outputs by their current stage; related research details appear in the sections above.

Work	Venue or format	Stage at 19 October 2026
Formation mechanism of blazar jets	Theoretical and Natural Science 19, 169–175	Published, 2023
Coherent strange-current contributions to X(2370) → ηηη	Physica Scripta	Submitted 9 October 2026
What Public AI Compute Records Can Establish: Evidence, Access and Comparison	International Journal of Technology, Policy and Management	Submitted 7 October 2026; Screening (last confirmed status)
Local modular-data convergence and locality/passband obstruction	Zenodo	Public version 2 preprint, 31 July 2026
Digit Dependence under Translation-Conjugated Cyclic Rotation	Research manuscript	Unpublished manuscript
Universal-prior/factorization no-go	Zenodo	Public 2026 preprint; later technical descendants described separately
Inference Rent in Generative AI	SSRN 6950759	Public working paper
Frontier AI and the Limits of Monopoly Self-Correction	SSRN 6950761	Public working paper
Camera, Player, Engine	SSRN 6950818	Public methodological preprint
Calibration diagnostics for ringdown overtone tests	Research manuscript	Working manuscript
Locality, Short Descriptions, and Machine-Dependent Ranking in a Finite Register	Research manuscript	In preparation
Dependence models and exact identification under uniform single deletion	Mathematical-statistics manuscript	Unpublished working manuscript
Positive population ambiguity after one deletion: composition crossover and exact critical growth	Mathematical-statistics manuscript	Unpublished working manuscript
An Unstated Constant Inside a Benchmark’s Metric	Physics-ML methods manuscript	Manuscript in preparation
An Invitation After Abundance	English-language book	Self-published, 2026
和平请柬	Chinese-language book	Self-published, 2026
Sovereign Scintillation Over Grand Fluctuation	Foundations-of-physics book manuscript	Unpublished
The Abundance That Isn’t; The Rent Machine	Essays	Drafts

Research practice, methods and languages

- **Research methods represented in the portfolio:** Monte Carlo/event simulation; timing and spectral analysis; amplitude and phase-space calculations; hierarchical inference and calibration; controlled computational experiments; exact finite-state and rational checks; operator-theoretic, combinatorial and statistical-identification arguments; source-traced documentary and comparative research.

- **Tools present in documented projects:** PYTHIA; HEASoft/NuSTARDAS, Powspec and XSPEC; Python scientific-computing and ML stacks; LALsuite and gravitational-wave/population-inference tools; LaTeX/RevTeX; web and local-interface tooling. This list describes the work represented here, without assigning an unverified proficiency rating or claiming every implementation was written unaided.
- **AI assistance and research contribution:** For the marked projects, AI assistance includes mathematical development and checking, code generation, computational analysis and manuscript drafting, where documented. Assistance in subsequent drafts and computations is distinct from the original research period; personal contributions and validation stages are specified separately in the project records.
- **Languages:** Chinese and English.

Detailed research CV · Updated 19 October 2026